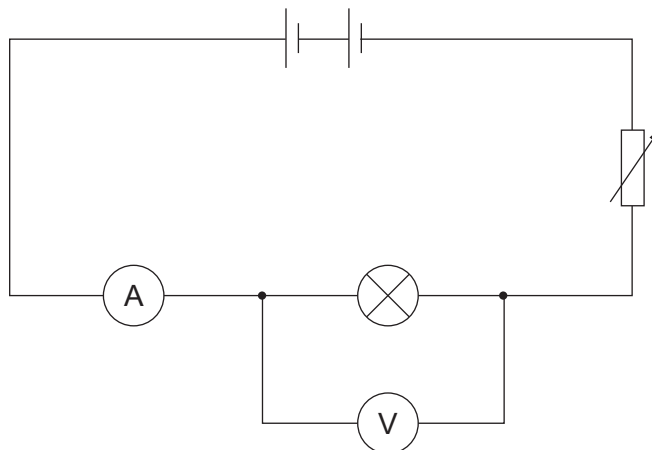


6. A student was investigating how current changes with voltage for various components. She used the following circuit to investigate a lamp.



She followed the method below.

1. Connect the circuit as shown in the diagram.
2. Set the variable resistor to give the lowest voltage and record the readings on the voltmeter and ammeter.
3. Alter the variable resistor to increase the voltage by 0.5 V.
4. Record the new readings on the voltmeter and ammeter.
5. Repeat steps 3 and 4.

(a) Some of the results are shown below.

Voltage (V)	Current (A)
0.0	0.00
2.5	0.65
3.5	0.90
4.5	1.16
5.0	1.20
5.5	1.26
6.5	1.36
8.0	1.46

(i) State the independent variable in this experiment.

[1]

.....

(ii) State the dependent variable in this experiment.

[1]

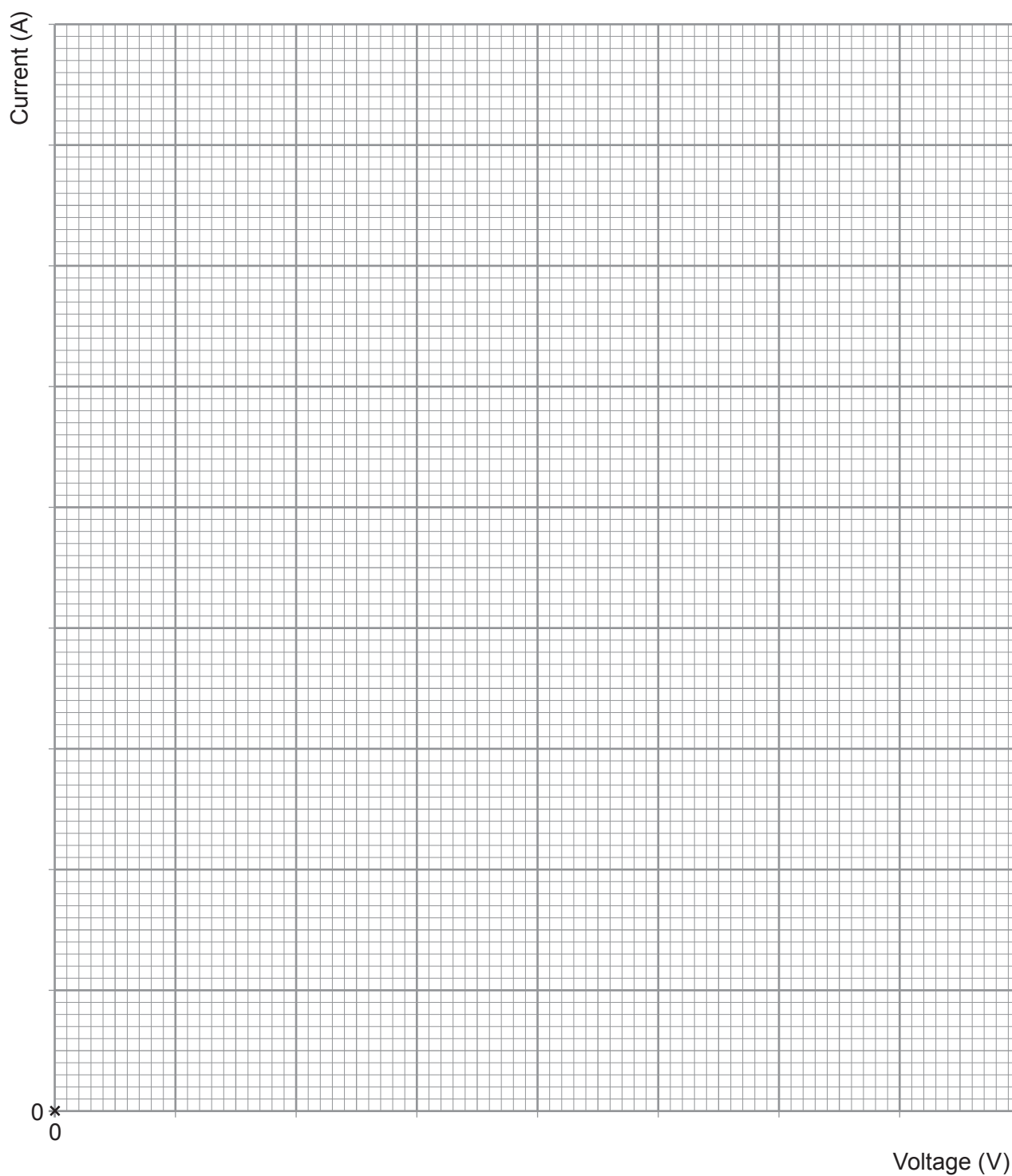
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(iii) Plot the results on the grid below and draw a suitable line.

[4]

Examiner
only



- (iv) Use your graph in (a)(iii) and the following equation

$$\text{voltage} = \text{current} \times \text{resistance}$$

to calculate the resistance of the lamp when the voltage is 7.2 V.

[3]

resistance = Ω

- (v) Explain whether the resistance of the lamp is constant throughout the experiment. [3]

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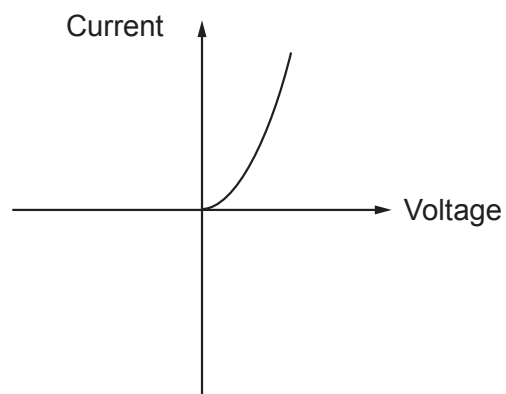
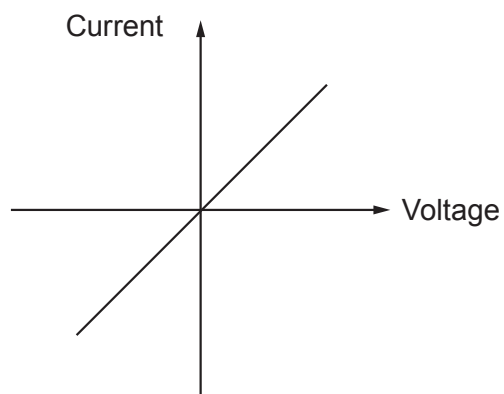
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- (b) The investigation was repeated with a resistor and then a diode. The graphs of the results are shown below.



Use the graphs to compare the variation in resistance for these components.

[4]

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END OF PAPER



2
1

4

6

1

0

$${}^1_1\text{H}$$

7 Li Lithium 3	9 Be Beryllium 4		
23 Na Sodium 11	24 Mg Magnesium 12		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89	

4 He Helium 2	19 F Fluorine 9	16 O Oxygen 8	14 N Nitrogen 7	12 C Carbon 6	11 B Boron 5
	35.5 Cl Chlorine 17	32 S Sulfur 16	31 P Phosphorus 15	28 Si Silicon 14	27 Al Aluminium 13
	80 Br Bromine 35	79 Se Selenium 34	75 As Arsenic 33	73 Ge Germanium 32	70 Ga Gallium 31
	127 I Iodine 53	128 Te Tellurium 52	122 Sb Antimony 51	119 Sn Tin 50	115 In Indium 49
	210 At Astatine 85	210 Po Polonium 84	209 Bi Bismuth 83	207 Pb Lead 82	204 Tl Thallium 81
222 Rn Radon 86	131 Xe Xenon 54	84 Kr Krypton 36	40 Ar Argon 18	20 Ne Neon 10	

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number